

(a) $\Delta x = \frac{\lambda}{2}$
 $\Delta \phi = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{2} = \pi$
 $\Delta \phi = \pi$
 $I_{res} = I_0 + 2I_0 + 2\sqrt{I_0 \cdot 2I_0} \cos \pi$
 $= 3I_0 - 2\sqrt{2}I_0$
 $= 0.2I_0$

(b) $\Delta \phi = \pi + \pi$
 $= 2\pi = 0$
 $I_{res} = I_0 + 2I_0 + 2\sqrt{I_0 \cdot 2I_0} \cos 0$
 $= 3I_0 + 2\sqrt{2}I_0$

(c) ω_1, ω_2 $\Delta \phi = (\omega_1 - \omega_2)t$
 $I_{res} = I_0 + 2I_0 + 2\sqrt{I_0 \cdot 2I_0} \cos((\omega_1 - \omega_2)t) \rightarrow [-1, 1]$
 $I_{Avg} = 3I_0$

A and B are two point sources of sound (of same frequency) and are kept at a separation. At a point P, the intensity of sound is observed to be I_0 when only source A is put on. With only B on the intensity is observed to be $2I_0$. The distance AP is higher than distance BP by half the wavelength of the sound. Find the intensity recorded at P with both sources on. Give your answer for following cases:

- The sources are coherent and in phase.
- The sources are coherent and 180° out of phase.
- The sources are incoherent.



Two sound sources oscillate in phase with a frequency of 100 Hz . At a point 1.74 m from one source and 1.16 m from the other, the amplitudes of sound from the two sources are A and $2A$ respectively. Calculate the amplitude of the resultant disturbance at the point. [Speed of sound in air is $v = 348 \text{ ms}^{-1}$]

$$\begin{aligned}\Delta\phi &= \frac{2\pi}{\lambda} \Delta x \\ &= \frac{2\pi}{3.48} \times 0.58 \\ &= \frac{\pi}{3}\end{aligned}$$

$$\lambda = \frac{v}{f} = \frac{348}{100} = 3.48$$

$$\begin{aligned}A_{\text{res}} &= \sqrt{A^2 + (2A)^2 + 2A(2A) \cos \frac{\pi}{3}} \\ &= \sqrt{5A^2 + 2A^2} = \sqrt{7}A\end{aligned}$$

$$\lambda_1 = \frac{18}{35} m = \frac{90}{175} m$$

$$\lambda_2 = \frac{90}{173} m$$

$$\lambda_1 < \lambda_2 \quad f = \frac{v}{\lambda}$$

$$f_1 > f_2$$

$$f_0 + 4 \quad | \quad f_0 - 4$$

$$f_1 - f_2 = 8$$

$$\frac{v}{\lambda_1} - \frac{v}{\lambda_2} = 8 = v \left\{ \frac{175}{90} - \frac{173}{90} \right\}$$

$$v = 360 m/s$$

$$f_1 = \frac{v}{\lambda} = \frac{360 \times 35}{18} = 700 Hz$$

$$f_0 = f_1 - 4 = 696 Hz$$

$$f_2 = f_1 - 8 = 692 Hz$$

Wavelength of two musical notes in air are $\frac{18}{35} m$ and $\left(\frac{90}{173}\right) m$. Each note produces four beats per second with a third note of frequency f_0 . Calculate the frequency f_0 .

$$V = \sqrt{\frac{\gamma R T}{M}} = \sqrt{\frac{\gamma R}{M}} T^{1/2}$$

$$V = k T^{1/2}$$

$$\ln V = \ln k + \frac{1}{2} \ln T$$

$$\frac{dV}{V} = 0 + \frac{1}{2} \frac{dT}{T}$$

$$dV = \frac{1}{2} \left(\frac{dT}{T} \right) V$$

$$\frac{dV}{dT} = \frac{1}{2} \cdot \frac{V}{T} = \frac{1}{2} \times \frac{331 \text{ m/s}}{273 \text{ K}}$$

$$= 0.6 \text{ m/s/K}$$

$$= 0.6 \text{ m sec}^{-1} \text{ Kelvin}^{-1}$$

Speed of sound in air is 331 ms^{-1} at 0°C . Prove that it increases at a rate of $0.6 \text{ ms}^{-1}^\circ\text{C}^{-1}$ for small temperature increase.

$$k = 155 \text{ m}^{-1}$$

$$\omega = 21000 \text{ rad/s}$$

$$v = \frac{\omega}{k} = \frac{21000}{155} = 1400 \text{ m/s}$$



$$A = \frac{\Delta P}{Bk} = \frac{3000 \pi \times 10^{-1} \text{ N/m}^2}{2 \times 10^9 \times 155 \text{ m}^{-1}} = 10^{-7} \text{ m} = 0.1 \mu\text{A}$$

$$I = \frac{(\Delta P)^2}{2 \rho v}$$

$$I = \frac{(3000 \pi)^2}{2 \times 1.02 \times 10^3 \times 1400} = 31.07 \text{ W/m}^2$$

$$I = \frac{P}{4\pi r^2}$$

$$P = I 4\pi r^2$$

$$= 31.07 \times 4\pi \times (10)^2 = 3.9 \times 10^4 \text{ W}$$

A point source of sound is located inside sea water. Bulk modulus of sea water is $B_w = 2.0 \times 10^9 \text{ N/m}^2$. A diver located at a distance of 10 m from the source registers a pressure amplitude of $\Delta P_0 = 3000 \pi \text{ N/m}^2$ and gives the equation of sound wave as

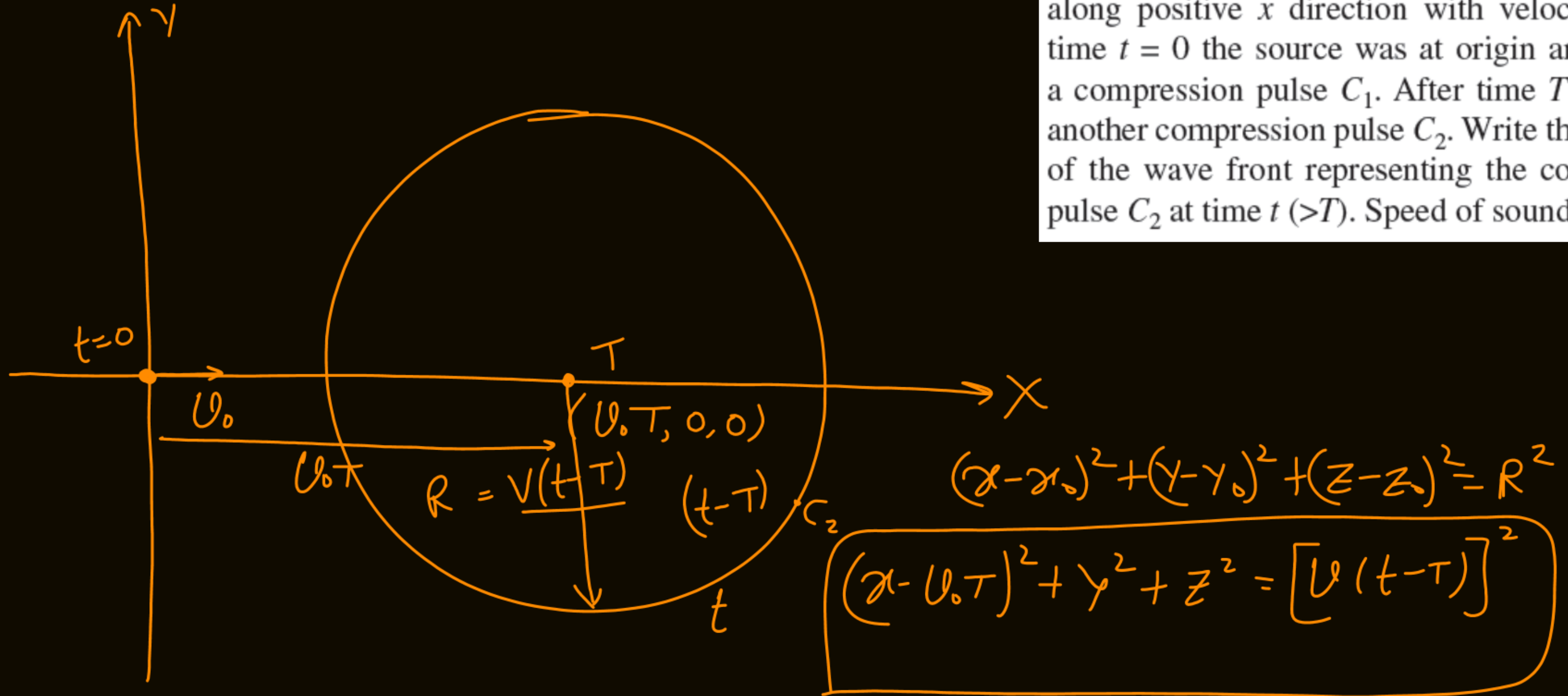
$y = A \sin(15 \pi x - 21000 \pi t)$, y and x are in meter and t is in second.

- Find the displacement amplitude of the sound wave at the location of the diver.
- Find the power of the sound source.

$$v = \sqrt{\frac{B}{\rho}}$$

$$\rho = \frac{B}{v^2} = \frac{2 \times 10^9}{(1400)^2} = 1.02 \times 10^3 \text{ kg/m}^3$$

A point source of sound is moving uniformly along positive x direction with velocity V_0 . At time $t = 0$ the source was at origin and emitted a compression pulse C_1 . After time T it emitted another compression pulse C_2 . Write the equation of the wave front representing the compression pulse C_2 at time $t (> T)$. Speed of sound is V .





$$330 \text{ m/s}$$

$$f_{\text{app}} = \frac{330 + 60}{330 - 30} \times 1200 \text{ Hz}$$

$$= \frac{39}{30} \times 1200 = 1560 \text{ Hz}$$

$$\lambda_1 = \frac{v_{\text{app}}}{f_{\text{app}}} = \frac{390 \text{ m/s}}{1560 \text{ Hz}} = \frac{1}{4} \text{ m}$$

$$f_2 = \frac{330 + 30}{330 - 60} \times 1560 = \frac{360}{270} \times 1560 \text{ Hz}$$

$$\lambda_2 = \frac{v_{\text{app}}}{f_{\text{app}}} = \frac{360 \times 270}{360 \times 1560} = \frac{27}{156} = \frac{9}{52}$$

$$\frac{\lambda_1}{\lambda_2} = \frac{1 \times 52}{4 \times 9}$$

$$= \frac{13}{9}$$

(i) A source of sound emits waves of frequency $f_0 = 1200 \text{ Hz}$. The source is travelling at a speed of $v_1 = 30 \text{ m/s}$ towards east. There is a large reflecting surface in front of the source which is travelling at a velocity of $v_2 = 60 \text{ m/s}$ towards west. Speed of sound in air is $v = 330 \text{ m/s}$.

(a) Find the number of waves arriving per second at the reflecting surface.

(b) Find the ratio of wavelength (λ_1) of sound in front of the source travelling towards the reflecting surface to the wavelength (λ_2) of sound in front of the source approaching it after getting reflected.

Two tuning forks produce 4 beats per second when they are sounded together. Now both the forks are moved towards the observer at same speed (u). The beat frequency now becomes 5 Hz. If the observer also begins to run with speed u towards both the forks, what beat frequency will he hear now?

$$f \quad f+4$$

C



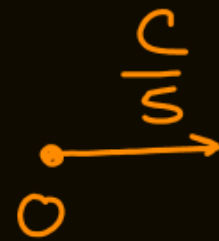
$$f_1 = \frac{C}{C-u} f$$

$$f_2 = \frac{C}{C-u} (f+4)$$

$$f_2 - f_1 = 5 = \frac{C \times 4}{C-u}$$

$$5C - 5u = 4C$$

$$C = 5u \Rightarrow \boxed{u = \frac{C}{5}}$$



$$f_1' = \frac{C + \frac{C}{5}}{C - \frac{C}{5}} f \Rightarrow \underline{\underline{f_1' = \frac{6}{4} f}}$$

$$f_2' = \frac{C + \frac{C}{5}}{C - \frac{C}{5}} (f+4)$$

$$= \underline{\underline{\frac{6}{4} (f+4)}} \quad \boxed{f_2' - f_1' = 6 \text{ Hz}}$$

$$L = \frac{5\lambda}{4}$$

$$= \frac{5}{4} \times \frac{3}{4}$$

$$= \frac{15}{16} \text{ m}$$



(ii) The air column in a pipe closed at one end is made to vibrate in its second overtone by a tuning fork of frequency 440 Hz. The speed of sound in air is 330 m/s. P_0 is mean pressure in the pipe and ΔP_0 is maximum amplitude of pressure variation. Neglect end correction.

(a) Find the length L of air column.

(b) What is amplitude of pressure variation at the middle of the column?

(c) What is maximum and minimum pressure at the closed end?

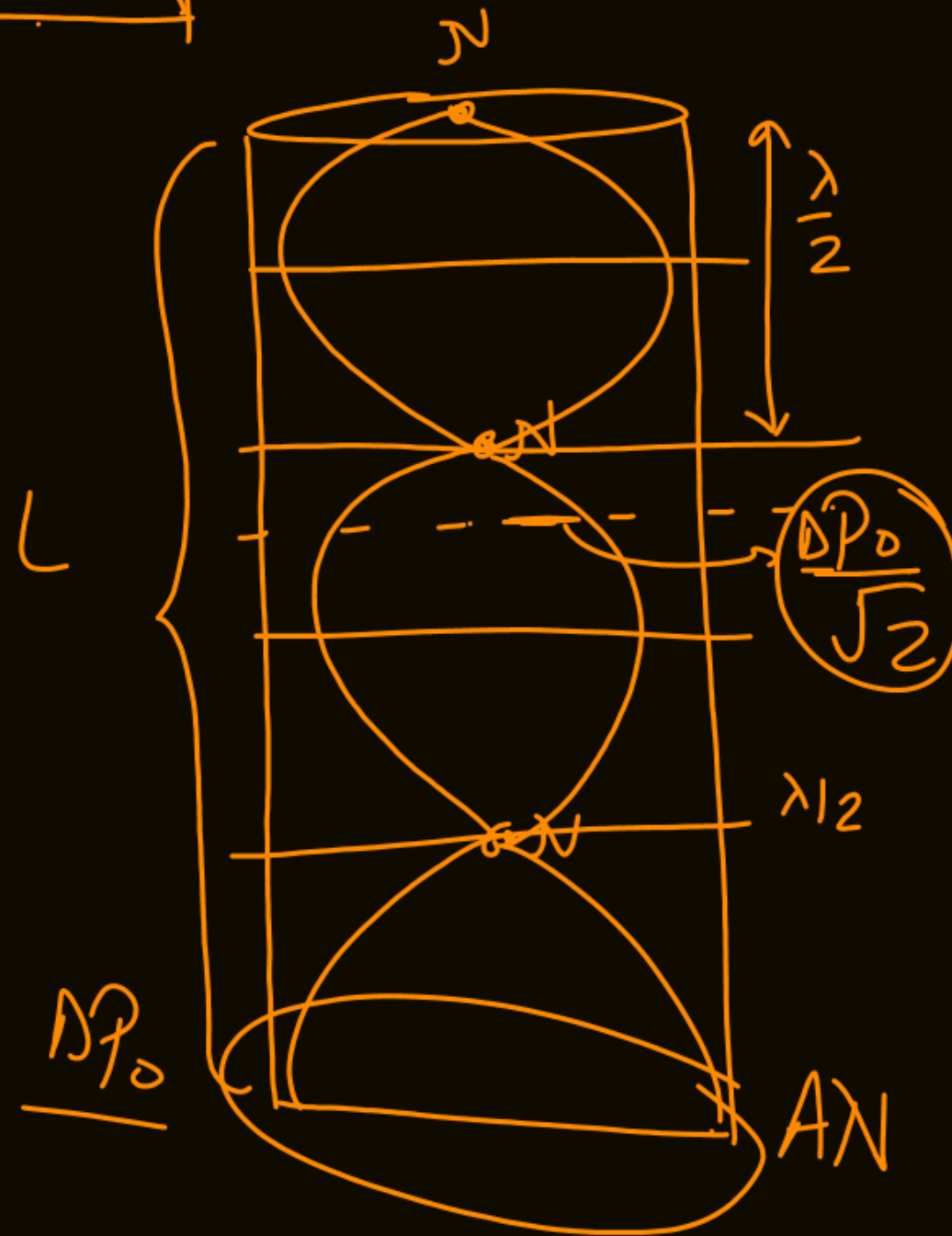
$$L = \frac{5\lambda}{4}$$

$$\lambda = 0.8L$$

$$L = 1.25\lambda$$

$$= (2.5) \frac{\lambda}{2}$$

$$P_0 - \Delta P_0 \quad P_0 + \Delta P_0$$



$$\Delta P = \Delta P_0 \sin kx$$

$$\Delta P = \Delta P_0 \sin \frac{2\pi}{\lambda} x$$

$$= (\Delta P_0) \sin \left(\frac{2\pi \times 5}{4} \cdot \frac{\lambda}{2} \right)$$

$$= (\Delta P_0) \sin \left(\frac{5\pi}{4} \right)$$

$$= -\frac{\Delta P_0}{\sqrt{2}}$$

① $M = \frac{n_1 M_1 + n_2 M_2}{n_1 + n_2}$

$$= \frac{3 \times 28 + 1 \times 32}{4}$$

$$= 29 \text{ g/mole}$$

$$\gamma = 1 + \frac{2}{(5)} = \frac{7}{5}$$

$$V = \sqrt{\frac{\frac{7}{5} \times \frac{25}{3} \times 300}{29 \times 10^{-3}}}$$

$$= \underline{347 \text{ m/s}}$$

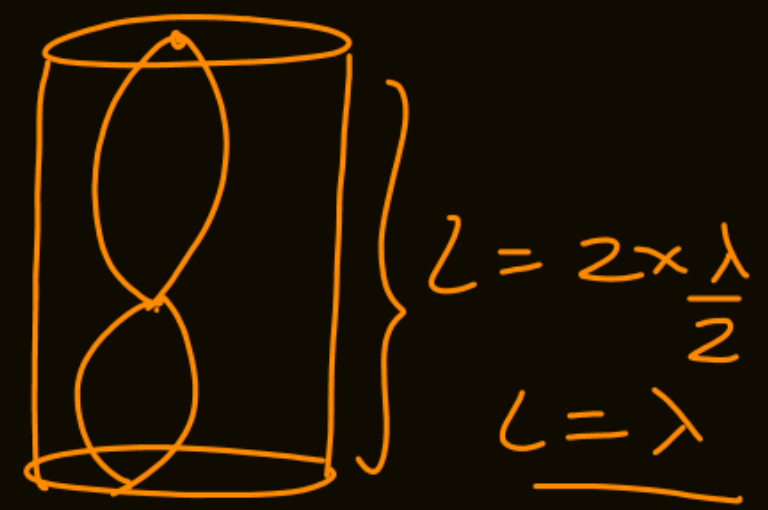
$$L = \lambda = \frac{347}{5000} \text{ m}$$

$$= 6.9 \times 10^{-2} \text{ m}$$

$$= 6.9 \text{ cm}$$

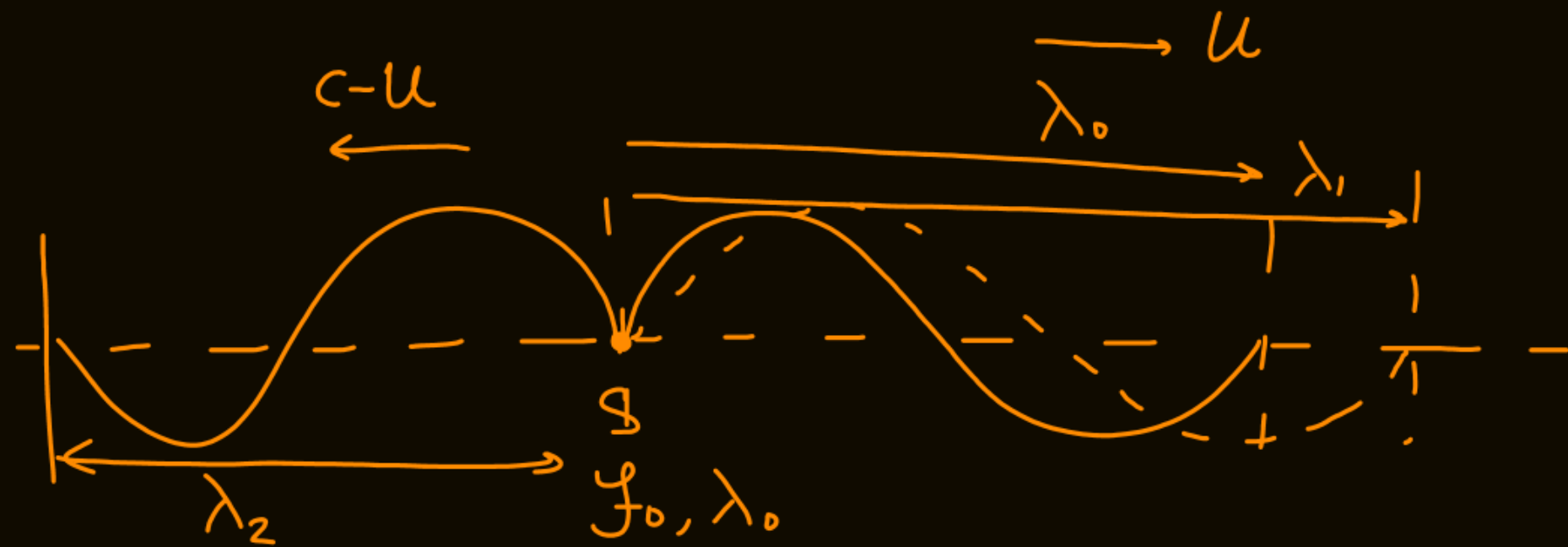


$$f = \frac{3V}{2L} = \frac{3 \times 347}{2 \times 6.9 \times 10^{-2}} = 7543 \text{ Hz}$$



Stationary wave of frequency 5 KHz is produced in a tube open at both ends and filled with air at 300 K. The tube is oscillating in its first overtone mode.

- (a) Find the length of the tube assuming that air contains only nitrogen and oxygen in molar ratio of 3 : 1.
- (b) What shall be the frequency of sound wave used so that the same tube oscillates in its second overtone mode?



A sound source emits waves of frequency f_0 and wavelength λ_0 in still air. When there is a wind blowing with speed u from left to right what will be wavelength of sound to the right of the source and to the left of the source.

$$C = \lambda_0 f_0$$

$$\lambda_0 = C \times \frac{1}{f_0}$$

$$\lambda_2 = (c - u) \times \frac{1}{f_0}$$

$$= \frac{c}{f_0} - \frac{u}{f_0}$$

$$\lambda_2 = \lambda_0 - \frac{u}{f_0}$$

$$\lambda_1 = (c + u) \times \frac{1}{f_0} = \frac{c}{f_0} + \frac{u}{f_0}$$

$$\lambda_1 = \lambda_0 + \frac{u}{f_0}$$